JASMEET SINGH

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GitHub | LinkedIn | Medium

ACADEMIC QUALIFICATION

B.E. Electronics and Communication — Bennett University (The Times Group) *CGPA: 7.97* | *Aug 2023*

WORK EXPERIENCE

Research Intern | *Bennett University, Greater Noida* *2023-2025*

- Conducted a comprehensive literature review of deep learning-based single image super-resolution and 3D pose estimation methods, analyzing state-of-the-art architectures and benchmarking datasets.
- Leveraged Python libraries such as Plotly, pythreejs, and Blender to develop interactive 3D visualizations for pose estimation data.
- Synthesized study findings through structured tables and visualizations, and evaluated model performance using PSNR and SSIM metrics.

Machine Learning Intern | *Feynn Labs Services* *Aug 2023 – Oct 2023*

- Performed customer segmentation using unsupervised ML techniques (clustering) and data analysis to derive actionable market insights.
- Developed business and financial models for products, including cost-benefit analysis and revenue projections to assess commercial viability.

ACADEMIC & PERSONAL PROJECTS

Autoregressive Image Generation — GitHub

- Built a VQ-VAE + GPT-style autoregressive transformer from scratch for CIFAR-10 image generation.
- Learned discrete visual representations via a 512-codebook VQ layer (vector quantization).
- Trained a 6-layer transformer for next-token prediction, reducing loss from ~6.2 to ~2.27.
- Applied straight-through estimation and multi-objective loss for stable and efficient convergence

Automating Test Case Generation with AI — GitHub

- Built RAG vector stores for regulatory compliance documents (FDA, ISO, GDPR) using Vertex AI RAG Engine to perform automated compliance checking on AI-generated test cases.
- Experimented with multiple RAG architectures including Agentic RAG with LangGraph and Graph RAG via LightRAG, and implemented a production-grade agentic RAG workflow using Vertex AI RAG Engine integrated with LangGraph.
- Deployed the RAG agent on Vertex AI Agent Engine and later exposed it as a scalable REST API on Google Cloud Run.

Text Summarization — GitHub

- Fine-tuned PEGASUS CNN/DailyMail model on the SAMSum dialogue dataset using Hugging Face Transformers, and evaluated performance with ROUGE metrics.
- Deployed a Streamlit application via CI/CD pipeline.

Neural Style Transfer with VGG-19 — HuggingFace Space

- Developed a neural style transfer application using VGG-19 in PyTorch, implementing style and content loss via Gram matrices and optimizing outputs through gradient-based methods.
- Deployed on Hugging Face Spaces with real-time GPU-enabled inference.

CERTIFICATIONS AND SPECIALIZATIONS

- PyTorch Specialization — DeepLearning.AI
- Machine Learning in Production — DeepLearning.AI
- Generative AI with LLMs — DeepLearning.AI
- Linear Algebra and Calculus for Data Science and Machine Learning — DeepLearning.AI